

Portfolio 1 - Variables and Expressions

Program 1: Student Information

This program prints my personal information such as my name, age, school program, and section.

Ask user for personal information

```
In [11]: name = input("Enter your name: ")
age = input("Enter your age: ")
program = input("Enter your program: ")
section = input("Enter your section: ")
```

Display student information

```
In [12]: print("----- STUDENT INFORMATION -----")
print("Name;", name)
print("Age:", age, "years old")
print("Program:", program)
print("Section:", section)
```

```
----- STUDENT INFORMATION -----
Name; Giemhel Buates
Age: 18 years old
Program: BS Chemical Engineering
Section: 1A
```

```
In [10]: print("Hola!", name)
```

```
Hola!, Giemhel Buates
```

This program demonstrates how variables and user input can be used to store and display data in a structured way.

Program 2: Student Average Grade

Write a program to prompt the user to input grades and calculate the average grade of a student based on major subjects for the first semester

Prompt the user to input grades for major subjects

```
In [15]: math10 = float(input("Enter grade for Math 10: "))
che110 = float(input("Enter grade for ChE 110: "))
chem11 = float(input("Enter grade for Chem 11: "))
chem11L = float(input("Enter grade for Chem 11L: "))
math11 = float(input("Enter grade for Math 11: "))
```

Compute Average Grade

```
In [19]: average = (math10+ che110 + chem11 + chem11L + math11)/5
```

Display Results

```
In [43]: print("\n----- STUDENT AVERAGE GRADE IN MAJOR SUBJECTS -----")
print("Math 10:", math10)
print("ChE 110:", che110)
print("Chem 11:", chem11)
print("Chem 11L:", chem11L)
print("Math 11:", math11)

print("\nAverage Grade:", average)
```

```
----- STUDENT AVERAGE GRADE IN MAJOR SUBJECTS -----
Math 10: 2.7
ChE 110: 1.6
Chem 11: 1.9
Chem 11L: 1.4
Math 11: 2.6
```

Average Grade: 2.04

This program demonstrates how variables and expressions are used to compute the average grade of a student based on major subjects. It show how input data can be processed using arithmetic operations to produce meaningful results by allowing the program to automatically compute values instead of doing manual calculations, making it more efficient and accurate.

Program 3: My School Transportation Expenses

Write a program that prompts the user to compute daily and weekly transportation based on school travel fares.

FARE PER ONE-WAY TRIP

Fare from home riding a jeepney to Legazpi

```
In [33]: fare_legazpi = 38
```

Fare riding a jeepney to Albay

```
In [34]: fare_albay = 13
```

Fare riding a tricycle to BUCENG

```
In [35]: fare_buceng = 10
```

Number of School days

```
In [36]: school_days = 6
```

COMPUTE ONE-WAY TRIP FARE (home → school)

```
In [37]: one_way_fare = fare_legazpi + fare_albay + fare_buceng
```

ROUND TRIP FARE (to school + going home)

```
In [38]: daily_fare = one_way_fare * 2
```

WEEKLY TRANSPORTATION EXPENSE

```
In [39]: weekly_expense = daily_fare * school_days
```

Display Results

```
In [44]: print("\n----- TRANSPORTATION EXPENSE SUMMARY -----")
print("One-way Fare:", one_way_fare)
print("Daily Fare (Round Trip):", daily_fare)
print("Weekly Transportation Expense:", weekly_expense)
```

----- TRANSPORTATION EXPENSE SUMMARY -----

One-way Fare: 61

Daily Fare (Round Trip): 122

Weekly Transportation Expense: 732

This program helps me estimate my daily and weekly transportation expenses for school.